A Leakage-Controlled Diagnostic of Protocol- and Calibration-Sensitive Multi-Task Gains in ADMET Prediction

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Abstract

We examine whether the relative proper-score performance of task-balanced hard-sharing multi-task (MTL) models in ADMET prediction depends on the evaluation protocol and the chemical novelty of test molecules. In a leakage-controlled diagnostic on eight binary ADMET endpoints from the Therapeutics Data Commons, we compare MTL with architecture-matched single-task models under global three-way molecule allocation (70%/10%/20% train/calibration/test) under grouped-random and scaffold-grouped protocols, with five split instances and three seeds; the final confirmatory plan was frozen before the corrected three-way rerun. The multi-task model assigns lower log loss on all eight endpoints under both protocols, and the benefit is larger under scaffold-grouped evaluation (protocol contrast $\mathbf{\Gamma = +0.081}$, split-instance block bootstrap 95% CI $\mathbf{[+0.048, +0.112]}$; positive in six of eight endpoints). Under the confirmatory design we find no systematic novelty dependence of the benefit under either protocol. A strict per-model temperature calibration—each temperature fitted on the same trained model's own calibration partition and applied to its own test partition—attenuates both the raw-probability advantage (post-calibration point estimates $\mathbf{-0.003}$ and $\mathbf{+0.003}$) and the protocol interaction (calibrated $\mathbf{\Gamma = +0.006}$), indicating that the observed gains primarily reflect temperature-correctable differences in raw logit scale. Compute-matched, label-permutation, and pooled-pretraining controls (under both protocols for the first two) indicate that the raw gain is not reproduced by additional single-task updates and does not require genuine auxiliary-label structure; the protocol interaction survives full label permutation. The positive direction of the protocol interaction was reproduced in a matched-scale graph neural network sensitivity analysis ($\mathbf{\Gamma_{GNN} = +0.026}$). These results indicate that reported multi-task advantages are conditional on evaluation protocol and should be accompanied by calibration diagnostics.

Scientific Contribution: We introduce a leakage-controlled framework for quantifying protocol-dependent multi-task versus single-task proper-score contrasts in ADMET prediction, and show that the multi-task log-loss advantage is larger under scaffold-grouped than grouped-random evaluation and is attenuated to near-zero point estimates by strict per-model temperature calibration.

Keywords: multi-task learning, ADMET prediction, evaluation protocol, calibration, benchmarking

1 Introduction

Multi-task learning (MTL) is widely used in ADMET (absorption, distribution, metabolism, excretion, and toxicity) modeling, and several studies report that multi-task models improve over matched single-task (STL) baselines Jiang et al. (2025); Shahid et al. (2025). At the same time, the molecular machine-learning community has accumulated evidence that evaluation protocols materially shape conclusions: out-of-distribution (OOD) split protocols such as scaffold or cluster splits change absolute performance and its rankings Fooladi et al. (2025); Guo et al. (2025), and recent benchmarks in neighboring fields have shown that strong model claims can depend on evaluation details Ahlmann-Eltze et al. (2025); Wei et al. (2026).

Existing split-protocol studies primarily ask how protocol changes a model's *absolute* performance. We instead ask how protocol changes the *paired MTL–STL contrast*. The second question does not follow from the first: a harder split could lower both models equally, widen the multi-task advantage, or reverse it. To our knowledge, prior work has not jointly characterized variation in the paired MTL–STL contrast across split protocol, target-task sample size, and continuous target-relative novelty while enforcing global molecule partitions.

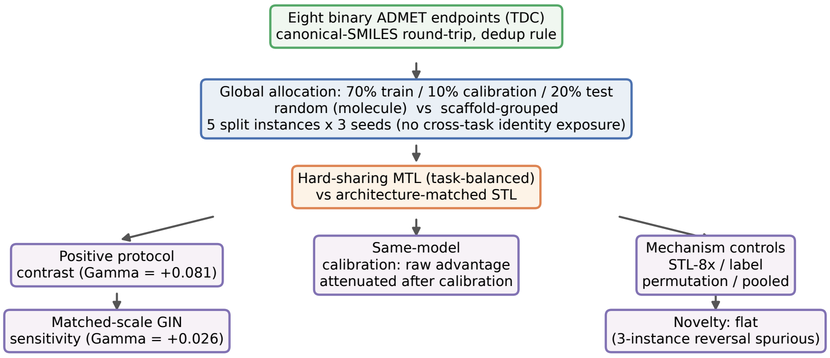

Fig. 1 Experimental framework. Eight binary ADMET endpoints are globally partitioned into train/calibration/test (no cross-task identity exposure) under grouped random and scaffold allocation; hard-sharing MTL is compared against architecture-matched STL via a molecule-level log-loss contrast. The analysis covers the protocol interaction Γ, strict same-model temperature calibration, mechanistic controls (compute-matched STL-8$\times$, label-permuted MTL, pooled pretraining), a matched-scale GIN sensitivity analysis, and the confirmatory novelty null result.

In this paper we report a leakage-controlled diagnostic whose final confirmatory plan was frozen before the corrected three-way rerun of task-balanced hard-sharing MTL on eight binary ADMET endpoints from the Therapeutics Data Commons Huang et al. (2021). The contributions are: (i) a paired MTL–STL protocol-contrast estimand evaluated under global molecule allocation with no cross-task identity exposure, under two valid protocols (grouped random and scaffold-grouped); (ii) a strict same-model calibration decomposition showing that the raw proper-score gain is attenuated to near-zero by per-model temperature rescaling; (iii) compute-matched (STL-8$\times$), label-permuted, and pooled-pretraining controls that narrow the mechanism to concurrent shared-trunk optimization with auxiliary inputs rather than semantic label transfer or additional target-task updates; and (iv) a matched-scale GNN sensitivity analysis together with a split-instance adequacy analysis. Endpoint-scale manipulation (downsampling) and the continuous novelty analysis are reported as secondary analyses.

The paper is organized as follows. Section 2 positions the work against prior studies of ADMET MTL, split protocols, and benchmark reliability. Section 3 defines the estimand hierarchy and the leakage-controlled design. Section 4 reports results along the three axes. Section 5 discusses implications and limitations.

2 Related Work

Multi-task ADMET modeling.

Multi-task architectures are common in ADMET prediction. MolP-PC Jiang et al. (2025) trains a multi-view, multi-task framework over 54 endpoints and reports that multi-task learning outperforms single-task training on 41 of 54 tasks; MTAN-ADMET Shahid et al. (2025) proposes an adaptive multi-task network over 24 endpoints with task-specific learning dynamics; industrial-scale studies report benefits of multi-task models on time-split data Walter et al. (2024). Reported gains vary across datasets and settings, and task balancing and endpoint size are themselves known to affect transfer in general MTL Kearnes et al. (2016). These studies report average or per-task performance under a single evaluation protocol and do not condition the reported contrast on protocol choice, endpoint scale, or novelty.

Split protocols and out-of-distribution evaluation.

A growing line of work examines how evaluation protocols change conclusions in molecular property prediction. Fooladi et al. (2025) compare 14 model classes across ten split protocols on eight datasets and find that scaffold splits are not uniformly harder than random splits while cluster-based splits are; Guo et al. (2025) reach similar conclusions on cancer cell lines and show that rank correlations between in-distribution and out-of-distribution performance collapse under cluster splits. Zheng et al. (2026) show that structural-frontier splits expose failures hidden by scaffold splits, and Guo and Ding (2026) find that model-size conclusions depend on protocol difficulty. These studies evaluate single-task models; the paired MTL–STL contrast is not their dependent variable, and none manipulates endpoint size or measures continuous target-relative novelty.

Multi-task transfer under realistic splits.

Studies at the intersection of MTL and split protocols are sparse and reach differing conclusions. Kearnes et al. (2016) observed that multi-task gains were modest and dataset-dependent under temporal splits in industrial ADMET data; Wu et al. (2026) found negative transfer in a two-task setting under scaffold splits, mitigated by safe scheduling; Walter et al. (2024) and CheMLT-F Mun and Fazli (2026) report multi-task benefits persisting under realistic splits. These results are not directly comparable because endpoint sets, architectures, metrics, and molecule-holdout schemes differ. A compact comparison table of these studies on six axes—architecture-matched STL baseline, split protocols, global molecule holdout, endpoint-size manipulation, continuous novelty, and probabilistic calibration—appears in the supplementary material; the table makes the gap concrete: no prior study jointly varies protocol, endpoint scale, and novelty while globally partitioning molecules and reporting calibration-level metrics.

Benchmark reliability and evaluation variability.

The molecular machine-learning community has been repeatedly reminded that evaluation choices can dominate model differences. In single-cell genomics, two major benchmarks reached contradictory conclusions about foundation-model perturbation prediction Ahlmann-Eltze et al. (2025); Wei et al. (2026), with the discrepancy traced to evaluation protocol details. Within molecular property prediction, rank correlations between in-distribution and out-of-distribution performance can collapse under cluster splits Guo et al. (2025), which implies that conclusions drawn from a single benchmark realization—including a single split instance and seed—should be treated cautiously. Our study follows this diagnostic tradition: we measure the conditions under which a design choice (task-balanced hard-sharing MTL) delivers or fails to deliver its reported advantage, with multiple split instances and seeds so that evaluation variability is quantified rather than assumed away.

3 Methods

3.1 Estimand hierarchy

For endpoint e, protocol p, split instance k, training seed r, and test set T_{epk}, let $\hat{p}^{\mathrm{STL}}_{iepkr}$ and $\hat{p}^{\mathrm{MTL}}_{iepkr}$ denote the predicted probabilities of the single-task and multi-task models for molecule i. We define the molecule-level log-loss contrast

$$d_{iepkr} \;=\; \ell(y_i, \hat{p}^{\mathrm{STL}}_{iepkr}) - \ell(y_i, \hat{p}^{\mathrm{MTL}}_{iepkr}), \tag{1}$$

where $\ell(y,p) = -y\log p - (1-y)\log(1-p)$; positive d favors the multi-task model. Endpoint-level and protocol-level aggregates are

$$\Delta_{epkr} = \frac{1}{|T_{epk}|}\sum_{i\in T_{epk}} d_{iepkr}, \qquad \Delta_p = \frac{1}{8}\sum_e \mathbb{E}_{k,r}[\Delta_{epkr}], \qquad \Gamma = \Delta_{\mathrm{scaffold}} - \Delta_{\mathrm{random}}. \tag{2}$$

Δ_p is the primary endpoint-macro estimand (endpoints weighted equally), and Γ is the protocol-contrast estimand. We treat the eight endpoints as a fixed target set; we do not claim a superpopulation of all ADMET endpoints. Log loss is a strictly proper measure of probabilistic accuracy; ranking discrimination (AUROC/AUPRC) is assessed separately.

3.2 Models and training

The multi-task model comprises a shared 1024–256–128 encoder (ReLU activations, dropout 0.2 after the first hidden layer) and eight independent affine heads mapping the 128-dimensional representation to one logit per endpoint. Model inputs are fixed-length binary fingerprints computed with RDKit's Morgan algorithm (`GetMorganFingerprintAsBitVect` RDKit contributors (2024)): radius 2, 1024 bits, default settings (`useChirality=False`, `useFeatures=False`), i.e., the standard non-chiral variant commonly denoted ECFP4; fingerprints were computed once per endpoint from the canonical SMILES and reused across all split instances and seeds. Each single-task model is an independently initialized encoder of the same architecture with one corresponding affine head; we therefore compare against *architecture-matched* single-task models, noting that capacity is matched only on the target-task pathway, not globally.

Training uses *equal task sampling with equal loss coefficients*. Each epoch consists of 64 MTL optimizer updates: eight batches of size $B = 512$ drawn with replacement from each of the eight tasks, each update using that task's unweighted binary cross-entropy; uniform scheduling gives equal task weighting in expectation. With 60 epochs, each task's head receives $U = 480$ updates, and the shared trunk receives 3,840 updates in total. Each single-task model receives $U = 480$ updates with $B = 512$ examples under the same sampler; collectively the eight STL models also receive 3,840 updates. Target-task exposure and optimizer-step count are therefore matched between MTL and STL; total examples processed differ (the MTL model additionally consumes auxiliary-task data). Optimization uses Adam ($\eta = 10^{-3}$, weight decay 10^{-4}) with the learning-rate schedule indexed by optimizer step. We scope all conclusions to this task-balanced hard-sharing architecture. As a sensitivity analysis, the protocol contrast was re-estimated with a graph neural network: a three-layer graph isomorphism network (GIN) with hidden width 64, sum pooling (`global_add_pool`), and per-endpoint linear heads, trained on molecular graphs with the same task-balanced schedule (Adam, $\eta = 10^{-3}$, batch size 128, 80 epochs, eight updates per task per epoch) under the same leakage-controlled design and both protocols, at the same scale as the MLP (five split instances $\times$ three seeds). All experiments ran under Python 3.10 with PyTorch 2.13.0

(CUDA 13.0 build) on a single NVIDIA RTX 4070 Ti SUPER GPU; RDKit 2023.09.6 was used for canonicalization, fingerprinting, and scaffold computation.

3.3 Mechanistic controls

Two additional training configurations isolate the mechanism of the contrast, both run under the random protocol at three split instances × three seeds with all other frozen settings unchanged. (i) *Label-permuted MTL*: for each target endpoint separately, the labels of all other seven endpoints are permuted within each auxiliary endpoint (fixed per-(instance, seed, endpoint) permutations), preserving input exposure, supervision count, and parameterization while removing genuine cross-task label information from the shared trunk; the target-task head still receives true labels. (ii) *Pooled pretrain + STL fine-tune*: a hard-sharing MTL run is used as a shared-trunk initialization, after which each endpoint's head is re-initialized and the full network is fine-tuned on that endpoint alone with the same task-balanced update schedule. These controls compare (i) label transfer against data exposure and (ii) joint-training gains against transferable-representation gains. A further compute-matched control (STL-8×) trains each single-task model on its own true data alone but with the multi-task trunk's full 3,840 optimizer updates (identical batch size, optimizer, and step-indexed schedule), so that optimization budget is matched while auxiliary-task input and labels are absent.

3.4 Leakage-controlled evaluation

A test molecule held out from endpoint e may still appear in the training partition of another endpoint, exposing its structure to the multi-task model through auxiliary labels. We control for this *cross-task identity exposure* by global molecule allocation: each molecule is assigned to a single global partition, and every endpoint inherits that partition. Evaluation is therefore inductive with respect to molecule identity across tasks: a test molecule cannot appear in any auxiliary endpoint's training partition. No cross-task identity exposure remains under this design. Molecule identity is canonicalized by RDKit canonical-SMILES round-trip (`MolFromSmiles`/`MolToSmiles`) under a fixed RDKit version, which retains defined stereochemistry in the canonical form. Per endpoint, rows that fail canonicalization are dropped, and rows sharing a canonical SMILES are deduplicated keeping the first occurrence; this is the frozen rule for duplicate and conflicting labels, applied identically in the development and confirmatory pipelines. Across the eight endpoints, 13 molecules carry conflicting labels (3 in hERG, 10 in BBB_Martins; none elsewhere); excluding all of them changes the endpoint-macro contrasts by less than 2×10^{-4} under both protocols and both raw and calibrated variants, so the deduplication rule does not drive any reported result. We evaluate two protocols under this scheme: grouped random allocation and Bemis–Murcko scaffold-grouped allocation Bemis and Murcko (1996) (with a defined policy for acyclic molecules and empty scaffolds). Five split instances and three training seeds are used; split and seed lists, and their hashes, are frozen. Each instance allocates 70% of the unique-molecule pool to training, 10% to calibration, and 20% to test, and this allocation is global: a molecule's partition is identical across all endpoints. The calibration partition exists solely for per-model temperature fitting (Section 3.7) and is disjoint from both training and test within each instance. The confirmatory dataset comprises 33,238 unique molecule–endpoint pairs from 17,972 unique canonical molecules; across the 5×3 (instance, seed) combinations this yields 148,008 test observations, with each (instance, seed) combination containing 9,814–9,930 unique molecule–endpoint pairs and no within-combination duplicates. Cross-instance overlap arises because different split instances draw different test partitions; the identity key is the same canonical SMILES used for global allocation.

3.5 Continuous target-relative novelty

For each test molecule we define *target-relative novelty* as one minus the maximum Tanimoto similarity (ECFP4 Morgan (1965), fixed RDKit version RDKit contributors (2024)) to any molecule in the target endpoint's training partition. Distance to auxiliary-task training molecules is deliberately excluded from the primary definition, so the axis measures distance from the endpoint's own training support; we preregister a secondary quantity for nearest-auxiliary-training similarity and its interaction with target-relative novelty, since a molecule far from target training but close to auxiliary training is precisely a candidate transfer case.

3.6 Endpoint scale and downsampling

Endpoint scale is the number of unique target-task training molecules in each split. With eight endpoints, an observational scale association is weak and confounded by endpoint identity; we therefore prespecify a downsampling experiment: large endpoints are subsampled to a fixed target-training size with a frozen sampler (nested subsets, class prevalence preserved, fixed validation and test partitions, auxiliary-task data unchanged, task-balanced schedule held constant). The defensible conclusion from this experiment is an effect of *target-label availability* within the examined endpoints under this training schedule, not a general "scale rather than identity" claim.

3.7 Calibration and discrimination decomposition

A Brier decomposition explains the Brier-score contrast, not the log-loss contrast. We therefore report (i) a Brier reliability–resolution decomposition as a descriptive complement, (ii) calibration intercept and slope, and (iii) AUROC/AUPRC contrasts. A finding of stable AUROC together with worse Brier reliability is reported as evidence *consistent with* a calibration difference, not as proof that the NLL penalty is primarily calibration.

Temperatures are fitted per run, protocol, model, and endpoint, and strictly within the same trained model. Global molecule allocation now assigns each molecule to one of three partitions: 70% training, 10% calibration, and 20% test (random protocol: molecule-level; scaffold protocol: scaffold-grouped). For each (split instance, training seed) run, one scalar temperature T is fitted by minimizing the mean log loss (NLL) on that same trained model's own calibration partition (disjoint from its training and test partitions; bounded scalar optimization, SciPy `minimize_scalar`, method `bounded`, $T \in [0.2, 5.0]$) and applied to that same model's own test partition; no temperature is shared across models. The calibrated molecule-level contrast is recomputed from temperature-scaled probabilities.

3.8 Uncertainty and preregistration

The fifteen split-seed combinations are not fifteen independent replicates: three training seeds share a test set, split instances reuse molecules, all endpoints share one MTL model, and a molecule can carry multiple endpoint labels. Confirmatory inference therefore uses paired contrasts with clustering by standardized molecule, distinguishing split variability from training variability; we report split-level and seed-level dispersion separately. The analysis plan—hypotheses, directional tests, split and seed hashes, estimands and weighting, exclusions, downsampling design, novelty model, multiplicity handling, and analysis code version—was frozen in an immutable record (repository commit 5c2cd1e; protocol/final_confirmatory_plan.md) before confirmatory computation on the leakage-controlled design; the development pass (a per-endpoint split subsequently identified as cross-task-exposed) informed the design and is reported separately as a contaminated stress test in the supplementary material, excluded from confirmatory statistics.

Endpoint	Test predictions	Unique molecules	Positive rate
hERG	1,938	443	0.69
AMES	21,933	4,914	0.55
BBB_Martins	5,907	1,322	0.76
Pgp_Broccatelli	3,747	832	0.53
CYP2C9_Veith	36,201	8,132	0.33
CYP2D6_Veith	39,246	8,829	0.19
CYP3A4_Veith	37,161	8,350	0.41
Bioavailability_Ma	1,875	416	0.77

Table 1 Dataset statistics. "Test predictions" counts molecule–endpoint test observations across 5 split instances × 3 seeds (within-(instance, seed) sets contain no duplicates; unique molecules across instances).

4 Results

We report results for eight binary ADMET endpoints from the Therapeutics Data Commons Huang et al. (2021): hERG, AMES, BBB penetration, P-glycoprotein inhibition, three cytochrome P450 inhibition tasks, and oral bioavailability (Table 1). All conclusions below are restricted to this endpoint set and to task-balanced hard-sharing MTL.

Across the leakage-controlled design (global molecule allocation, five split instances, three seeds), the molecule-level log-loss contrast Δ is positive for every endpoint under both protocols: multi-task models assign lower log loss than their architecture-matched single-task counterparts in all 8×2 protocol–endpoint configurations.

4.1 Protocol interaction and overall effect

The endpoint-macro mean contrast is $\Delta_{\text{random}} = +0.321$ under grouped random allocation and $\Delta_{\text{scaffold}} = +0.402$ under Bemis–Murcko scaffold-grouped allocation, with a protocol contrast of $\Gamma = \Delta_{\text{scaffold}} - \Delta_{\text{random}} = +0.081$ (paired $t_7 = 2.65$, $p = 0.033$). Because the eight endpoints and five split instances are fixed in this study, uncertainty in Γ has three distinct components, which we report separately. Conditional on the observed split instances and training runs, molecule-level resampling gives a narrow interval (95% CI $[+0.079, +0.083]$); this answers only "how much sampling noise among molecules," not variability across instance realizations. A two-level block bootstrap that resamples split instances first (then aggregates endpoints within them) gives 95% CI $[+0.048, +0.112]$ ($P(\Gamma > 0) = 1.00$), covering split-instance variability. An endpoint-resample bootstrap (resampling the eight Γ_e values) gives 95% CI $[+0.026, +0.138]$ ($P(\Gamma > 0) = 0.998$), covering endpoint heterogeneity; the across-endpoint standard deviation of Γ_e is 0.088. All three intervals exclude zero: the positive protocol contrast remained stable under molecule-, split-instance-, and endpoint-level resampling within this benchmark panel. The endpoint-resample interval is reported as a sensitivity analysis; the eight endpoints are a fixed target set, so it does not imply an ADMET-endpoint population. The per-endpoint contrasts Γ_e are positive in six of eight endpoints (Figure 2, panel B); the interaction is concentrated among the smaller endpoints in this panel (hERG, AMES, BBB, Bioavailability, each $\Gamma_e \approx +0.16$), while the two largest CYP endpoints show $\Gamma_e \approx 0$ or slightly negative (CYP2C9 -0.012, CYP3A4 -0.028). A leave-one-endpoint-out analysis leaves the interaction in the range $+0.069$ to $+0.096$, so no single endpoint carries the interaction. We present these as concordance and sensitivity analyses for the eight fixed endpoints; the endpoint-dispersion interval does not imply an ADMET endpoint population, and the paired test is a fixed-set diagnostic, not a superpopulation claim. The multi-task benefit is therefore *larger under the scaffold protocol* for the smaller endpoints, and the paired MTL–STL contrast is protocol-dependent: moving from grouped-random to scaffold-grouped evaluation need not shrink the observed MTL advantage. We report the contrast on these eight fixed endpoints; we do not claim an ADMET-wide population effect.

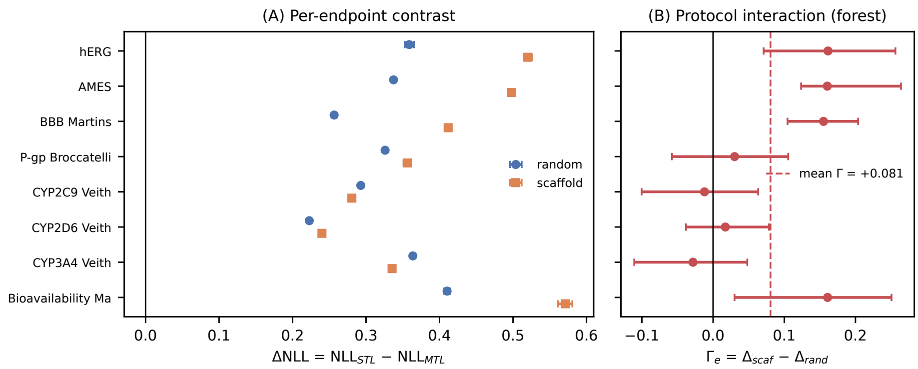

Fig. 2 (A) Per-endpoint log-loss contrast $\Delta = \mathrm{NLL}_{\mathrm{STL}} - \mathrm{NLL}_{\mathrm{MTL}}$ (positive values favor MTL) under grouped random (circles) and scaffold-grouped (squares) allocation, with molecule-level cluster-bootstrap 95% intervals. (B) Per-endpoint protocol contrasts $\Gamma_e = \Delta_{\mathrm{scaf}} - \Delta_{\mathrm{rand}}$ (forest) with endpoint-macro mean (dashed). Six of eight endpoints show positive Γ_e; leave-one-endpoint-out range +0.069 to +0.096.

4.2 Endpoint-scale moderation

Observationally, the contrast is largest for the smallest endpoints under both protocols (e.g., Bioavailability_Ma +0.41; hERG +0.36 under random) and smallest for the largest ones (CYP2D6 +0.22), and the same ordering holds under scaffold allocation (Figure 2). A prespecified downsampling intervention subsampled the three CYP endpoints to 2,000 target-task training molecules under the current 70/10/20 design: the contrast increased in all three (CYP2C9 +0.28 → +0.40; CYP2D6 +0.24 → +0.38; CYP3A4 +0.34 → +0.53), consistent with shared representations helping more when target-task labels are scarcer; because the trunk is shared, non-CYP endpoints also moved (mixed direction), so we do not claim a causal "scale rather than identity" mechanism (full details in Supporting Information S4).

4.3 Continuous novelty analysis and window test

Under the confirmatory design (five split instances, both protocols) the contrast shows no systematic dependence on target-relative novelty under either protocol (Figure 3): within-endpoint novelty slopes straddle zero for most endpoints under both protocols, and a hierarchical interaction model yields a detectable but negligible

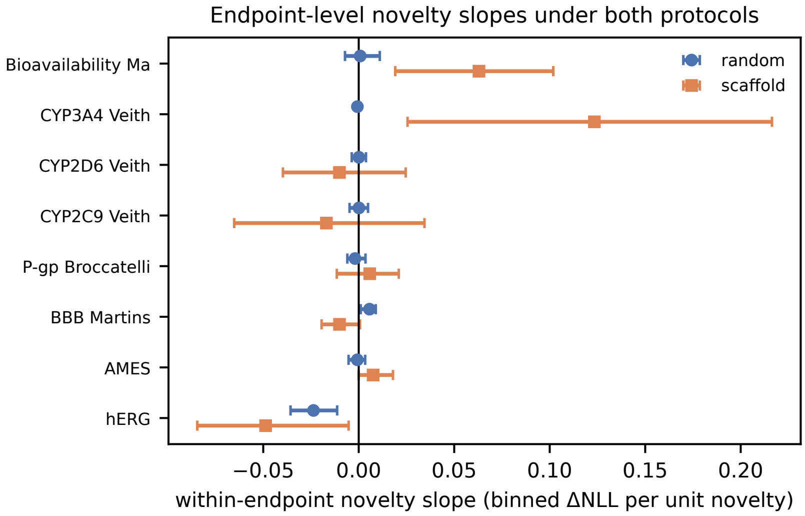

Fig. 3 Novelty slopes under the confirmatory design. Within-endpoint slopes of the binned contrast on target-relative novelty under random (circles) and scaffold (squares) allocation, with unit-level bootstrap intervals. Slopes straddle zero for most endpoints under both protocols; an apparent reversal seen in a three-instance development analysis did not survive five instances (Supporting Information Figure S1).

effect ($\hat{\beta} = -0.006$; 95% CI $[-0.010, -0.001]$)—a full-unit novelty increase changes the contrast by less than 2% of the random-protocol macro contrast. The preregistered transfer-window hypothesis is therefore not supported under either protocol. A three-instance development analysis had suggested a striking reversal of the novelty dependence; this pattern did not survive the five-instance confirmatory design (Supporting Information Figure S1), which we report as a methodological caution about group-sampling instability with few instances.

4.4 Calibration/discrimination decomposition and robustness

The contrast decomposes into raw-probability and calibration components. Under the random protocol, Brier scores and reliability components improve for all eight endpoints while AUROC is essentially unchanged; under the scaffold protocol, Brier and reliability improve everywhere and AUROC improves in six of eight endpoints (e.g.,

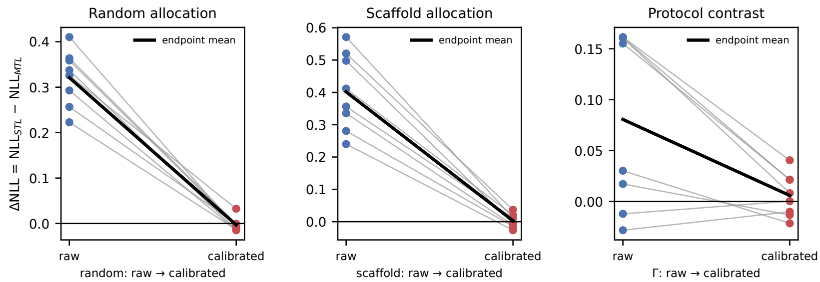

Fig. 4 Temperature-calibration contrast. Mean Δ under both protocols, raw versus strict per-model temperature calibration: the raw advantage (+0.321 random, +0.402 scaffold) attenuates to approximately zero, and the protocol contrast Γ from +0.081 to +0.006 (points: per-endpoint values).

hERG 0.76 → 0.78). To test whether the log-loss advantage reflects a durable calibration difference or merely better raw probability estimates, we fitted a per-endpoint temperature T Guo et al. (2017) under the strict per-model design: for each (split instance, training seed) run, protocol, model, and endpoint, one scalar temperature was fitted by minimizing the mean log loss on that same trained model's own calibration partition (disjoint from its training and test partitions) and applied to that same model's own test partition; no temperature is shared across models. After this calibration, the mean contrast collapses from +0.321 to −0.003 under the random protocol and from +0.402 to +0.003 under the scaffold protocol, and the calibrated protocol contrast is $\Gamma_{\text{cal}} = +0.006$ (per-endpoint calibrated contrasts positive in five of eight endpoints; paired $t_7 = 0.80$, $p = 0.45$). The uncalibrated gains—including the protocol interaction ($\Gamma_{\text{raw}} = +0.081$ vs. $\Gamma_{\text{cal}} = +0.006$)—are therefore attenuated to near-zero point estimates by per-model temperature rescaling (Figure 4), consistent with differences in raw logit scale rather than with ranking information or a calibration advantage that survives recalibration; we report point estimates and note that a formal equivalence analysis against a practical margin is needed before claiming elimination. Class-conditional analysis shows the contrast is carried predominantly by positive-class molecules (+0.46 vs. +0.20 for negatives under random allocation).

4.5 Partition-geometry controls

Because novelty is defined relative to each endpoint's own training partition, the scaffold protocol could in principle produce its larger contrast merely by shifting test molecules to different novelty levels. Reweighting the random-protocol test molecules so their novelty distribution matches the scaffold protocol's leaves the random-protocol contrast essentially unchanged (+0.3004 vs. +0.3006): the scaffold advantage is not explained by the between-protocol novelty shift (full details in Supporting Information).

4.6 Decision-level implications

Proper-score contrasts do not directly translate to practical selection behavior, so we compared MTL and STL on top-k compound prioritization for the three toxicity-relevant endpoints, computed separately within each of the 15 (instance, seed) runs to avoid pooling repeated molecules (ranking is invariant to temperature scaling, so the comparison is unaffected by calibration). Per-run differences are small: across the 18 protocol–endpoint–k combinations ($k \in \{50, 100, 200\}$), the mean difference in true toxicants captured in the top k is between -1 and $+2$ molecules, with MTL matching or exceeding STL in the majority of runs for 16 of 18 combinations, and the conclusion does not depend on the illustrative cost ratio ($c \in \{2, 5, 10\}$). We therefore do not claim a general prioritization advantage; the analysis is endpoint-local and descriptive (full results in Supporting Information S6).

4.7 What drives the contrast: mechanistic controls

The MTL and STL systems differ in representation sharing but also in data exposure and optimizer budget: the MTL shared trunk receives 3,840 optimizer updates (eight tasks $\times$ 480 per-task updates) versus 480 for each STL trunk, and the MTL model processes auxiliary-task molecules the STL model never sees. Three controls under the random protocol isolate these channels (three split instances × three seeds for the label and pretraining controls; five instances × three seeds for the budget control; all other frozen settings unchanged). First, a *compute-matched STL* baseline (STL-8×): each endpoint is trained on its own true data alone but with the MTL trunk's 3,840 updates (identical batch size, optimizer, and schedule). STL-8× is *worse* than standard STL (−0.096 log-loss units; negative in all 15 runs): the multi-task advantage was not reproduced by extending single-task training to the matched 3,840-update budget under the frozen optimizer and learning-rate schedule. (We report the fixed final-step checkpoints; no test-based early stopping was used, so the control answers "same final update budget," not "independently tuned best STL.") Second, a *label-permuted MTL* model: for each target endpoint separately, the labels of all *other* seven endpoints are permuted within each endpoint (fixed per-(instance, seed, endpoint) permutations; identical input exposure, supervision count, and parameterization), so the shared trunk cannot extract genuine cross-task label information while the target head still receives true labels. This fully permuted control leaves the contrast essentially unchanged (+0.293 vs. +0.300 for standard MTL; +0.387 vs. STL-8×), indicating that the raw gain does not require genuine auxiliary-label information and is not reproduced by additional target-task updates. Third, a shared trunk pretrained jointly on all endpoints and then fine-tuned per endpoint (head re-initialized, full-network fine-tuning) retains only a small fraction of the contrast (+0.045, versus +0.300 for hard-sharing joint training), showing that most of the gain requires the joint training objective itself rather than a transferable shared representation alone (Figure 5). The same two decisive controls were repeated under the scaffold protocol (three split instances × three seeds): label-permuted MTL leaves the scaffold-protocol contrast essentially unchanged (+0.441 vs. +0.407 for standard MTL), and the protocol interaction survives full label permutation ($\Gamma_{\text{permuted}} = +0.112$ vs. $\Gamma_{\text{real}} = +0.096$ at matched three-instance scale), so the protocol-sensitive gain is not attributable

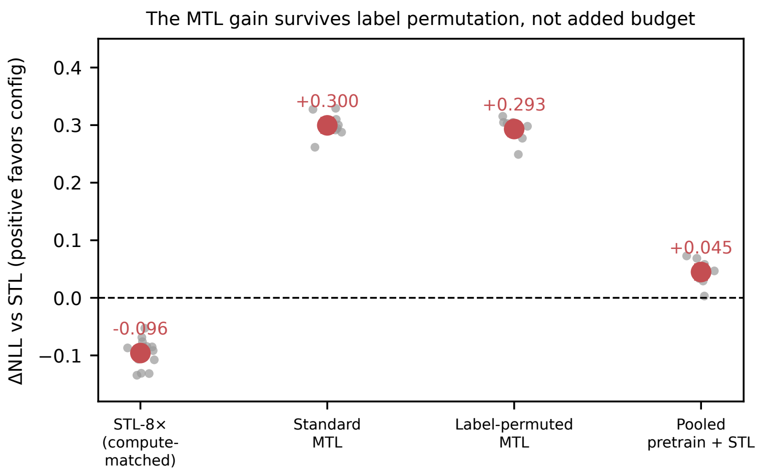

Fig. 5 Mechanistic controls (random protocol; points: split–seed estimates, error bars: 95% intervals across runs). Compute-matched STL-8× (8× optimizer budget on single-task data) is *worse* than standard STL (−0.096), while standard MTL improves log loss over STL by +0.300, fully permuting all auxiliary labels (target-specific) leaves the contrast essentially unchanged, and pooled pretraining followed by per-endpoint fine-tuning retains only +0.045. Under grouped-random evaluation, the controls narrow the explanation to concurrent shared-trunk optimization with auxiliary inputs rather than semantic label transfer or additional target-task updates.

to semantic cross-task label transfer either. STL-8× is again worse than STL under the scaffold protocol (−0.125), confirming that additional single-task updates do not reproduce the gain in either protocol. Together, the controls narrow the plausible explanation to effects specific to concurrent shared-trunk optimization with auxiliary inputs—including auxiliary-input exposure and gradient-noise regularization—rather than semantic cross-task label transfer or additional target-task updates; they do not uniquely identify a single mechanism. An earlier control that permuted only a fixed subset of auxiliary endpoints produced unstable negative transfer; we attribute that instability to optimization sensitivity of the partially permuted configuration and report only the fully permuted design here. These controls are endpoint-local diagnostics of the main finding.

4.8 Cross-architecture robustness

To test whether the protocol interaction is specific to fingerprint MLPs, we repeated the core comparison with a three-layer graph isomorphism network (GIN) on molecular graphs, under the same leakage-controlled design at matched scale (five split instances, three seeds, architecture-matched STL). The positive direction of the protocol contrast is reproduced: $\Delta_{\text{random}} \approx 0$ (-0.001) and $\Delta_{\text{scaffold}} = +0.025$, giving $\Gamma_{\text{GNN}} = +0.026$ (unit-level bootstrap 95% CI $[+0.007, +0.046]$; the interval excludes zero), although its magnitude and endpoint pattern differ from the fingerprint MLP (positive Γ_e in four of eight endpoints, driven by hERG and Bioavailability). We read the GNN result as supporting the direction of the interaction in a second representation and model family rather than establishing representation independence. An earlier GNN run with a smaller training budget showed apparent negative transfer; this was an underfitting artifact that disappeared once the training budget matched the MLP configuration.

Endpoint-effect ordering across split instances is not stable (mean pairwise Kendall $\tau = 0.17$, range $[-0.29, 0.57]$), indicating that conclusions about which endpoints benefit most should not be drawn from a single split realization. Full per-endpoint estimates (contrast, AUROC, AUPRC, Brier, reliability, unique molecule counts) appear in Table S2 of the Supporting Information.

5 Discussion

5.1 What the results show

The results qualify, rather than overturn, the commonly reported multi-task advantage in ADMET modeling Jiang et al. (2025); Shahid et al. (2025): under leakage-controlled evaluation, task-balanced hard-sharing MTL provides better raw probability estimates on all eight endpoints and both protocols. The headline finding is the protocol interaction: the paired MTL–STL proper-score contrast is +0.321 under random allocation and +0.402 under scaffold allocation ($\Gamma = +0.081$; paired $t_7 = 2.65$, $p = 0.033$), and the positive contrast was robust to molecule-, split-instance-, and endpoint-level resampling (split-instance block bootstrap 95% CI [+0.048, +0.112]); leave-one-endpoint-out range +0.069 to +0.096, positive in six of eight endpoints and concentrated among the smaller endpoints. Under the confirmatory five-instance design we found no systematic novelty dependence of the benefit under either protocol. The raw-probability advantage—and the protocol interaction itself—does not survive strict per-model temperature calibration, in which each temperature is fitted on the same trained model's own calibration partition and applied to its own test partition (calibrated $\Gamma = +0.006$, paired $p = 0.45$), so the gains reflect temperature-correctable differences in raw logit scale rather than ranking information or a durable calibration advantage.

5.2 Relation to prior split-protocol and MTL findings

The protocol-dependence finding extends single-task protocol studies Fooladi et al. (2025); Guo et al. (2025); Zheng et al. (2026) to the multi-task contrast, and it helps resolve the apparent tension between studies reporting multi-task benefits under realistic splits Walter et al. (2024); Mun and Fazli (2026) and studies reporting negative transfer Wu et al. (2026); Kearnes et al. (2016): the direction and magnitude of the benefit depend on protocol and on the metric (proper score versus ranking). The temperature-calibration result adds a methodological caution: raw-probability advantages that disappear under recalibration should be reported as such, and calibration diagnostics should accompany proper-score comparisons. The decision-level analysis revealed only small and heterogeneous ranking differences: MTL matched or exceeded STL in most run-level top-k comparisons, but two protocol–endpoint–k combinations favored STL more often, so we do not infer a general compound-prioritization advantage.

5.3 Methodological lessons

Two results in this study were design-sensitive in ways that reward explicit reporting. First, a three-instance scaffold analysis showed an apparent reversal of the novelty dependence that disappeared under the five-instance design; the GNN sensitivity analysis similarly showed apparent negative transfer before the training budget matched the MLP configuration. Both episodes illustrate that grouped-allocation studies should verify instance and budget adequacy before interpreting pattern-level claims, and they motivate the frozen protocol and the separation of development from confirmatory analyses adopted here. Second, the strict per-model calibration design—fitting each temperature on the same model's own calibration partition—yields the same qualitative conclusion as an earlier cross-instance calibration shortcut, but the strict design removes a legitimate reviewer objection to the shortcut; we recommend the strict design whenever calibration is part of the evidence.

5.4 Limitations

The endpoint set is limited to eight binary TDC endpoints and one task-balanced hard-sharing fingerprint MLP architecture; other architectures, weighting schemes, and endpoint families may behave differently. The downsampling manipulation increased the contrast consistently across the three CYP endpoints, but because the multi-task model shares its trunk, the intervention also altered the auxiliary endpoints; the result therefore supports target-label availability as a moderator within this training configuration but does not isolate a general causal sample-size mechanism. Endpoint-effect ordering is unstable across split instances, and with eight endpoints the protocol interaction, while supported by the paired test and bootstrap intervals at all three resampling levels, rests on a small number of endpoint-level observations; we do not claim an ADMET-wide population effect. The temperature-calibration analysis is a point-estimate attenuation analysis; a formal equivalence analysis against a practical margin would be needed to claim elimination, and the calibrated point estimates should be read as approximately zero rather than as proof of exact zero. The top-k decision analysis is endpoint-local and descriptive. We treat all findings as benchmark-local.

6 Conclusion

We presented a leakage-controlled diagnostic with a frozen confirmatory plan of task-balanced hard-sharing multi-task ADMET models. The multi-task proper-score contrast is positive on all eight endpoints under both protocols, larger under scaffold-grouped evaluation ($\Gamma = +0.081$; split-instance block bootstrap 95% CI $[+0.048, +0.112]$; positive in six of eight endpoints), with the interval robust to molecule- and endpoint-level resampling (sensitivity analyses in the Results), and attributable to better raw probability estimates that do not survive strict per-model temperature calibration (calibrated $\Gamma = +0.006$, $p = 0.45$). Under the confirmatory five-instance design we found no systematic novelty dependence of the benefit under either protocol; an apparent reversal in a three-instance development analysis did not survive the larger design.

Compute-matched and label-permutation controls (under both protocols) indicate that the raw gain is not reproduced by additional single-task updates and does not require genuine auxiliary-label structure—the protocol interaction survives full label permutation—and the positive direction of the protocol interaction was reproduced in a matched-scale GNN sensitivity analysis ($\Gamma_{\mathrm{GNN}} = +0.026$, bootstrap 95% CI $[+0.007, +0.046]$). These results suggest that multi-task proper-score advantages in ADMET modeling should be reported conditionally on evaluation protocol and accompanied by calibration diagnostics, and that instance count in grouped-allocation studies should be large enough to stabilize group sampling. Natural next steps are to extend the diagnostic to larger endpoint panels, alternative multi-task formulations, and additional OOD protocols such as cluster or temporal splits.

Declarations

Availability of data and materials

The dataset(s) supporting the conclusions of this article are available in the Therapeutics Data Commons repository (https://tdcommons.ai/), endpoint identifiers: hERG, AMES, BBB_Martins, Pgp_Broccatelli, CYP2C9_Veith, CYP2D6_Veith, CYP3A4_Veith, Bioavailability_Ma. The analysis code, global molecule-partition split manifests, prediction tables, temperature-calibration module, and the full analysis protocol are available in the GitHub repository https://github.com/Functionhx/ADMET-MTL-Diagnostic (MIT license; frozen submission snapshot: release v1.0.0). Archived version with persistent DOI: https://doi.org/10.5281/zenodo.21772295.

Competing interests

The authors declare that they have no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Authors' contributions

Y.F. conceived the study, designed the experiments, performed the analysis, and wrote the manuscript.

Acknowledgements

Not applicable.

Use of AI tools

AI-assisted tools were used for writing support and code assistance during the preparation of this manuscript; all experiments, analyses, and conclusions were performed and verified by the author.

List of abbreviations

- ADMET: Absorption, distribution, metabolism, excretion, and toxicity
- AUPRC: Area under the precision-recall curve
- AUROC: Area under the receiver operating characteristic curve
- ECFP4: Extended connectivity fingerprint, radius 4 equivalent (Morgan radius 2)
- GIN: Graph isomorphism network
- MTL: Multi-task learning
- NLL: Negative log likelihood
- OOD: Out of distribution
- STL: Single-task learning
- TDC: Therapeutics Data Commons

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